

OOPS ASSIGNMENT – 09

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FUNCTION TEMPLATE:

1.

```
#include <iostream>
```

```
using namespace std;
```

```
template <typename T>
```

```
void swapValues(T &a, T &b) {
```

```
    T temp = a;
```

```
    a = b;
```

```
    b = temp;
```

```
}
```

```
int main() {
```

```
    int x = 5, y = 10;
```

```
    swapValues(x, y);
```

```
    cout << x << " " << y << endl;
```

```
    float a = 1.5, b = 2.5;
```

```
    swapValues(a, b);
```

```
    cout << a << " " << b << endl;
```

```
    return 0;
```

```
}
```

```

(base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash8.cpp -o trash8 && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash8
10 5
2.5 1.5
(base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

2.

```
#include <iostream>
```

```
using namespace std;
```

```
template <typename T>
```

```
T findMin(T arr[], int n) {
```

```
    T minVal = arr[0];
```

```
    for(int i = 1; i < n; i++) {
```

```
        if(arr[i] < minVal)
```

```
            minVal = arr[i];
```

```
    }
```

```
    return minVal;
```

```
}
```

```
int main() {
```

```
    int a[] = {5, 2, 9, 1, 7};
```

```
    cout << findMin(a, 5) << endl;
```

```
    float b[] = {2.5, 0.5, 3.1, 1.2};
```

```
    cout << findMin(b, 4) << endl;
```

```
    return 0;
```

```
}
```

```

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1
0.5
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```

3.

```
#include <iostream>
```

```
using namespace std;
```

```
template <typename T>
```

```
void bubbleSort(T arr[], int n) {
```

```
    for(int i = 0; i < n - 1; i++) {
```

```
        for(int j = 0; j < n - i - 1; j++) {
```

```
            if(arr[j] > arr[j + 1]) {
```

```
                T temp = arr[j];
```

```
                arr[j] = arr[j + 1];
```

```
                arr[j + 1] = temp;
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
int main() {
```

```
    int a[] = {5, 2, 9, 1, 7};
```

```
    int n = 5;
```

```
    bubbleSort(a, n);
```

```
    for(int i = 0; i < n; i++)
```

```
        cout << a[i] << " ";
```

```
    return 0;
```

```
}
```

```
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1 2 5 7 9
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```

4.

```
#include <iostream>
```

```
using namespace std;
```

```
template <typename T>
```

```
int linearSearch(T arr[], int n, T key) {
```

```
    for(int i = 0; i < n; i++) {
```

```
        if(arr[i] == key)
```

```
            return i;
```

```
    }
```

```
    return -1;
```

```
}
```

```
int main() {
```

```
    int a[] = {5, 2, 9, 1, 7};
```

```
    int n = 5;
```

```
    cout << linearSearch(a, n, 9) << endl;
```

```
    return 0;
```

```
}
```

```
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2
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```

5.

```
#include <iostream>
```

```
using namespace std;
```

```
template <typename T>
void process(T a) {
    cout << a << endl;
}
```

```
template <typename T>
void process(T a, T b) {
    cout << a << " " << b << endl;
}
```

```
template <typename T, typename U>
void process(T a, U b) {
    cout << a << " " << b << endl;
}
```

```
int main() {
    process(5);
    process(5, 10);
    process(5, 2.5);
    return 0;
}
```



```
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5
5 10
5 2.5
○ (base) vihaan@Vihaans-Macbook 10.GRAPHs %
```

CLASS TEMPLATE:

1.

```
#include <iostream>
```

```
using namespace std;
```

```
template <typename T>
```

```
class Stack {
```

```
    T arr[100];
```

```
    int top;
```

```
public:
```

```
    Stack() { top = -1; }
```

```
    void push(T x) {
```

```
        if(top == 99) return;
```

```
        arr[++top] = x;
```

```
    }
```

```
    void pop() {
```

```
        if(top == -1) return;
```

```
        top--;
```

```
    }
```

```
    T peek() {
```

```
        return arr[top];
```

```
    }
```

```
};
```

```
int main() {
```

```
    Stack<int> s;
```

```
    s.push(10);
```

```

s.push(20);

cout << s.peek() << endl;

s.pop();

cout << s.peek() << endl;

return 0;

}

```



```

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20
10
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```

2.

```

#include <iostream>

using namespace std;

```

```

template <typename T>

```

```

class Queue {

    T arr[100];

    int front, rear;

```

```

public:

```

```

    Queue() {

        front = 0;

        rear = -1;

    }

```

```

    void enqueue(T x) {

        if(rear == 99) return;

        arr[++rear] = x;

    }

```

```

void dequeue() {
    if(front > rear) return;
    front++;
}

T peek() {
    return arr[front];
}
};

```

```

int main() {
    Queue<int> q;
    q.enqueue(10);
    q.enqueue(20);
    cout << q.peek() << endl;
    q.dequeue();
    cout << q.peek() << endl;
    return 0;
}

```

```

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10
20
(base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

3.

```

#include <iostream>
using namespace std;

```

```

template <typename T1, typename T2>
class Pair {

```



```

    T1 first;

    T2 second;

public:
    Pair(T1 a, T2 b) {
        first = a;
        second = b;
    }

    void display() {
        cout << first << " " << second << endl;
    }
};

int main() {
    Pair<int, float> p(10, 2.5);
    p.display();
    return 0;
}

```

```

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10 2.5
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```

4.

```
#include <iostream>
```

```
using namespace std;
```

```
template <typename T>
```

```
class Arithmetic {
```

```
    T a, b;
```

```
public:
```

```

Arithmetic(T x, T y) {
    a = x;
    b = y;
}

T add() { return a + b; }
T subtract() { return a - b; }
T multiply() { return a * b; }
T divide() { return a / b; }
};

int main() {
    Arithmetic<int> obj(10, 5);
    cout << obj.add() << endl;
    cout << obj.subtract() << endl;
    cout << obj.multiply() << endl;
    cout << obj.divide() << endl;
    return 0;
}

```



```

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15
5
50
2
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```

5.

```

#include <iostream>

using namespace std;

template <typename T>
class Array {

```

```

T arr[100];

int n;

public:

void input(int size) {
    n = size;
    for(int i = 0; i < n; i++)
        cin >> arr[i];
}

void display() {
    for(int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
}

};

```

```

int main() {
    Array<int> a;
    int n;
    cin >> n;
    a.input(n);
    a.display();
    return 0;
}

```

```

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2
3
4
3 4
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```